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Course Title: System Analysis and Design

Section: 01

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Title:

Technology Used in Small Businesses

Submitted to:

Raihan Kibria Senior

Lecturer

Department of Computer Science and Engineering (CSE)

ULAB School of Science & Engineering

University of Liberal Arts Bangladesh

Submitted by:

Raju Ahmed (241014131)

MD.Iftekharul Alam Sikder(241014054)

Nasrin Sultana Muskan(241014019)

Nayem Hasan (242014196)

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1. Introduction

This project will present the concept of the "Small Business Technology System" that aims to streamline and automate business operations. Using Context Diagram, DFD Level 0, and DFD Level 1, we will demonstrate the process of how the system operates in terms of data flow among customers, owners, managers, and suppliers. The technology is specifically concerned with automating activities such as order processing, inventory management, secure transactions, and business reports. With all these activities being automated through a technological tool, the Small Business Technology System makes it easier for small businesses to eliminate mistakes, ensure fast service, and improve their interactions with both employees and customers.

2. Stakeholder Analysis

1. Customer

The Customer is the key beneficiary of the system. They interact with the system through ordering products and submitting payments. The main duty of customers when using the system is to give order details and pay for the items purchased. The system gives them confirmations for their orders and payments to facilitate easy buying.

2. Business Owner

The Business Owner is the one who monitors the whole activity of the company. He does not handle small activities like checking the inventory level. The system offers him reports known as “Business Reports.” This report shows him the performance of the business, which he uses to make business decisions.

3. Manager

The Manager manages the daily activities in the business. He checks whether the inventory levels are good enough. In case of need for more products from the supplier, the manager uses the system to request the products he wants from the supplier.

4. Suppliers

The suppliers are outside agencies that supply the company with its goods. Suppliers communicate with the system via the inventory module. Whenever there is a need for restocking, the system sends an inventory order to the suppliers. This enables the suppliers to supply the items needed and update the system regarding the new stock level.

3. System Vision Document

Problem Statement

A significant number of small enterprises are faced with manual problems, including keeping records of stock manually, selling their stock manually, making entries manually, and accepting orders from clients manually using the phone or text messages. Such a manual process is often associated with errors in calculations, missing details of transactions, and delays in supplying the needs of the clients. Lack of a proper system makes it difficult for the owners to monitor their businesses efficiently.

System Description

The "Small Business Technology System" refers to an IT infrastructure that automates processes of small businesses' functioning. In other words, it provides users with access to products offered by the company and places orders online. The system automatically monitors and controls stocks to ensure timely replenishment of inventory. Moreover, it processes transactions and saves information about each purchase into the digital database.

Strategic Objectives

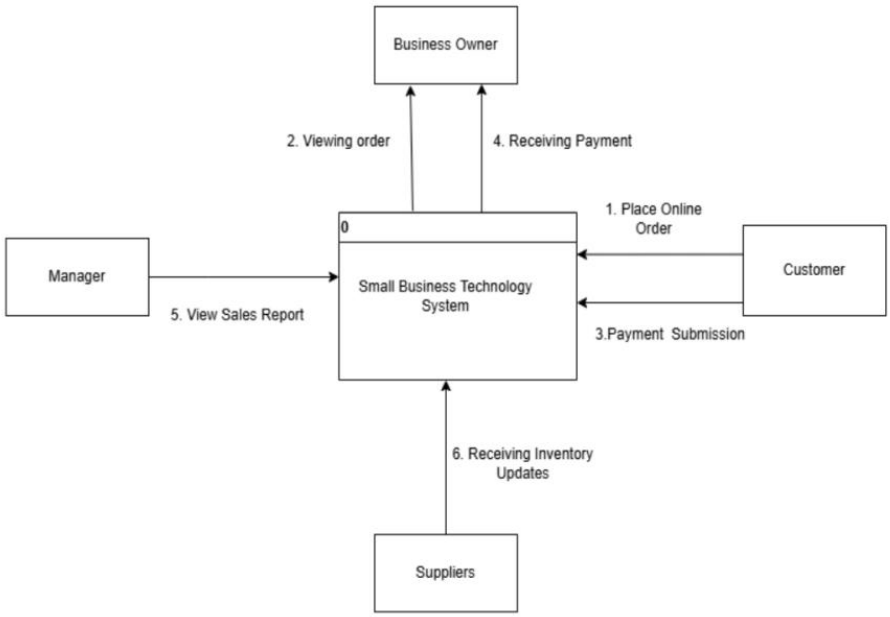
The primary purpose of such a system is to increase efficiency and minimize errors in small business processes. This task can be achieved by automating order and stock management operations, which will reduce the workload on employees. In addition, one of the objectives pursued by implementing this system is to collect analytical data that will serve as a basis for decision-making in terms of the development of the company.

Expected Benefits

After implementation, there will be various advantages. The customers will benefit from the ease of buying products, and as such, they will feel satisfied. The business will save time since there will be no filling of papers, and all data will be entered automatically. Also, there will be better inventory management such that the business will not lack any item. Finally, the business owner will always know the total amount of sales made.

4. System Architecture and Diagrams (Items 1-9)

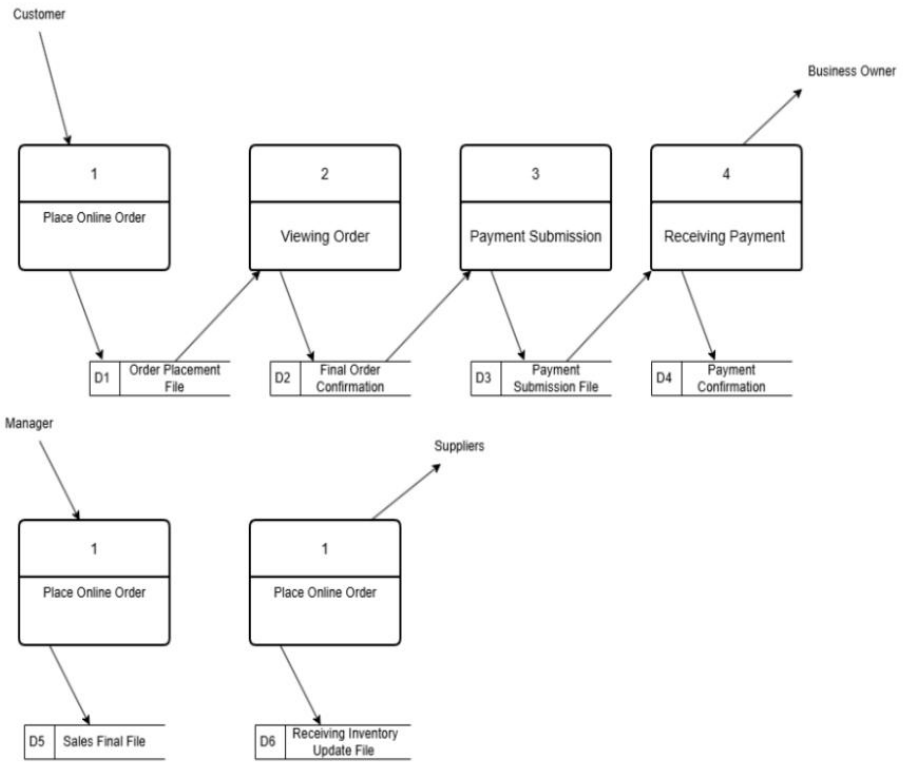
4.1 Context Diagram:



4.2 DFD LEVEL 0:

DFD 0:

1. Place Online Order
2. Viewing Order
3. Payment Submission
4. Receiving Payment
5. View Sales Report
6. Receiving Inventory Updates



4.3 DFD LEVEL 1:

Process Breakdown Diagram:

1. Place Online Order:

    1.1 Select Product

    1.2 Submit Order

2. Viewing Order:

    2.1 Retrieve Order Details

    2.2 Display Order Status

3. Payment Submission:

    3.1 Enter Payment Info

    3.2 Confirm Payment

4. Receiving Payment:

    4.1 Verify Payment

    4.2 Update Payment Status

5. View Sales Report:

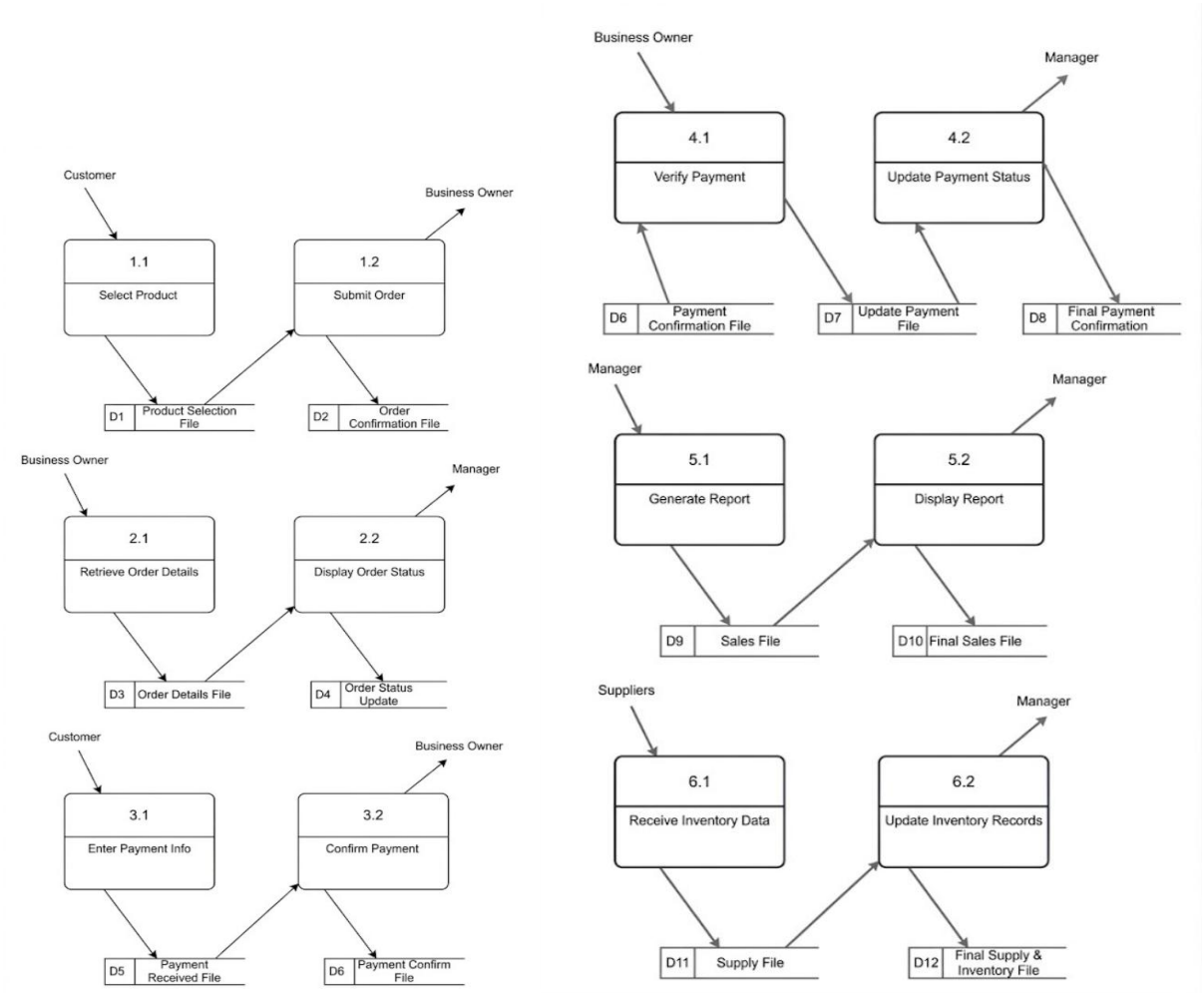
    5.1 Generate Report

    5.2 Display Report

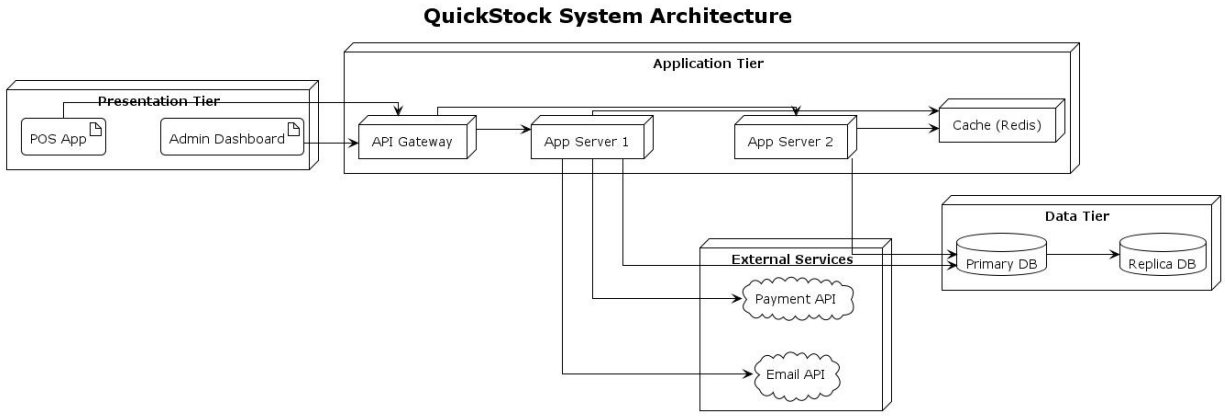
6. Receiving Inventory Updates:

    6.1 Receive Inventory Data

    6.2 Update Inventory Records

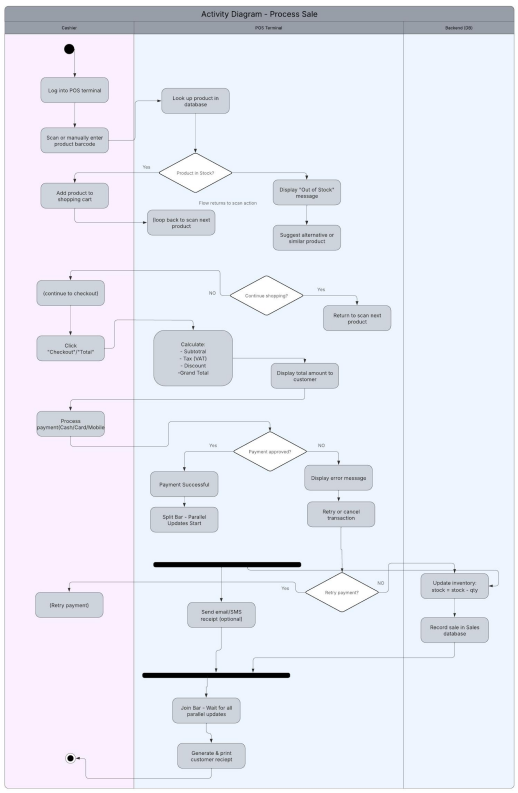


4.4 System Architecture Configuration Diagram:

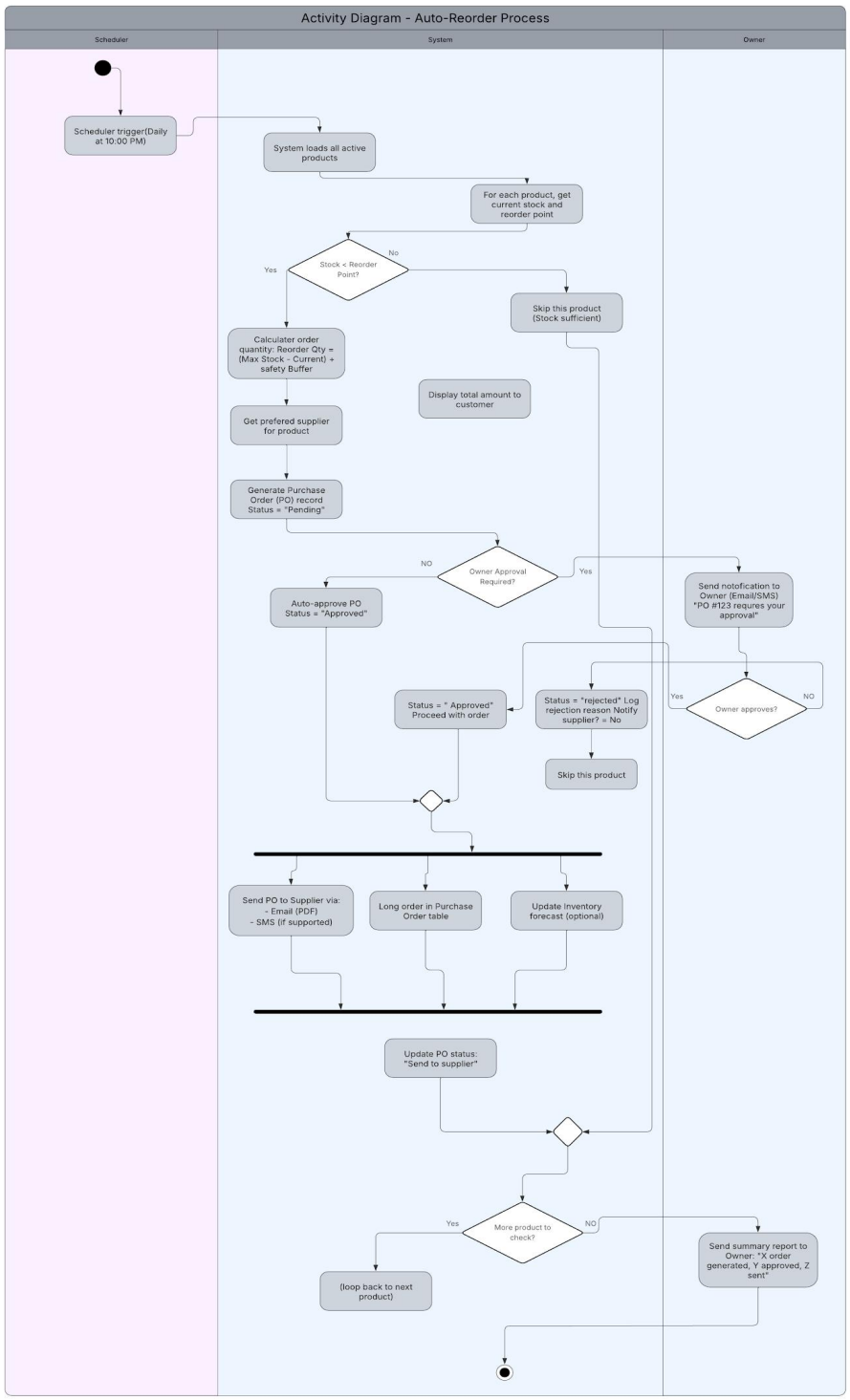


4.5 Activity Diagrams:

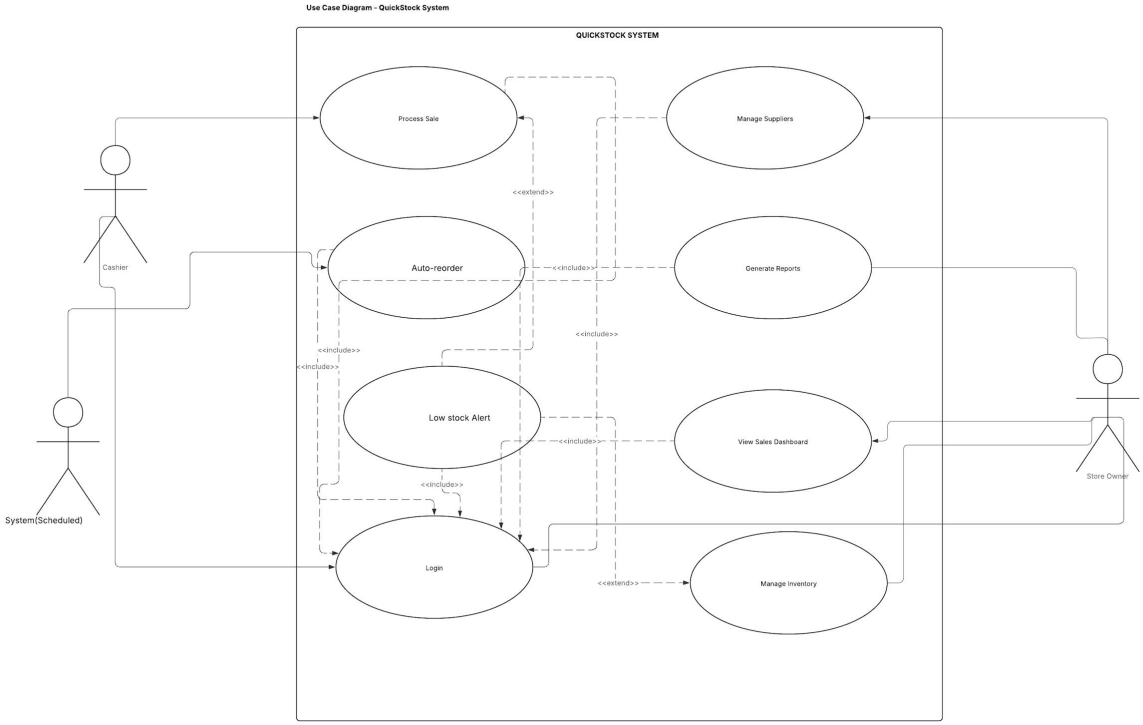
Activity Diagram for Process Sale



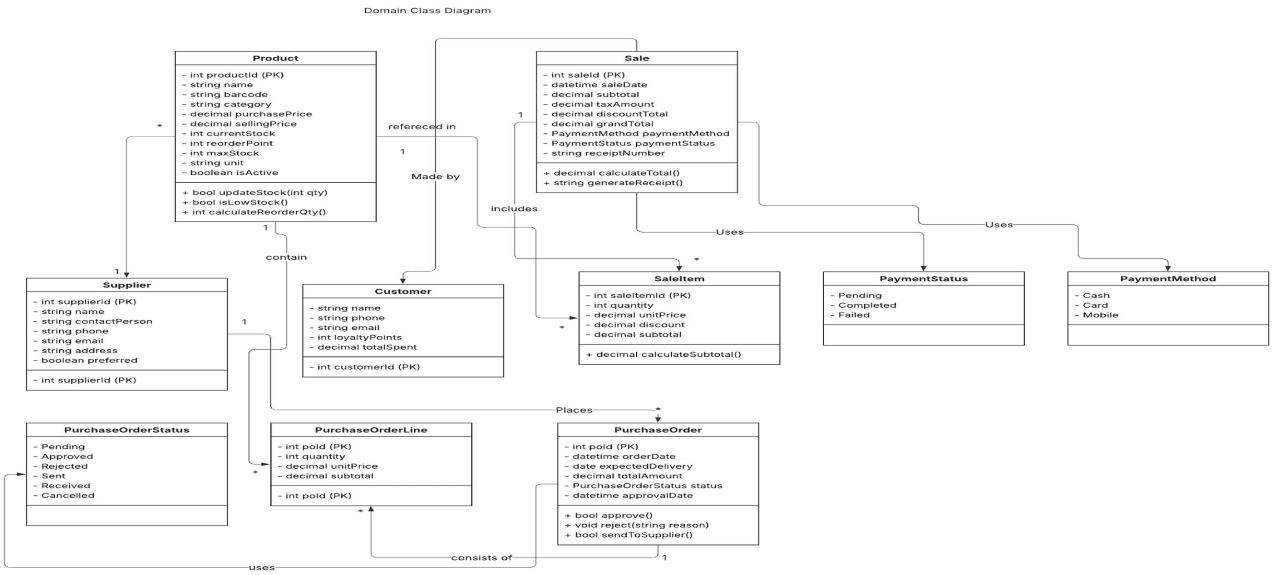
Activity Diagram for Auto-Reorder



4.6 Use Case Diagram

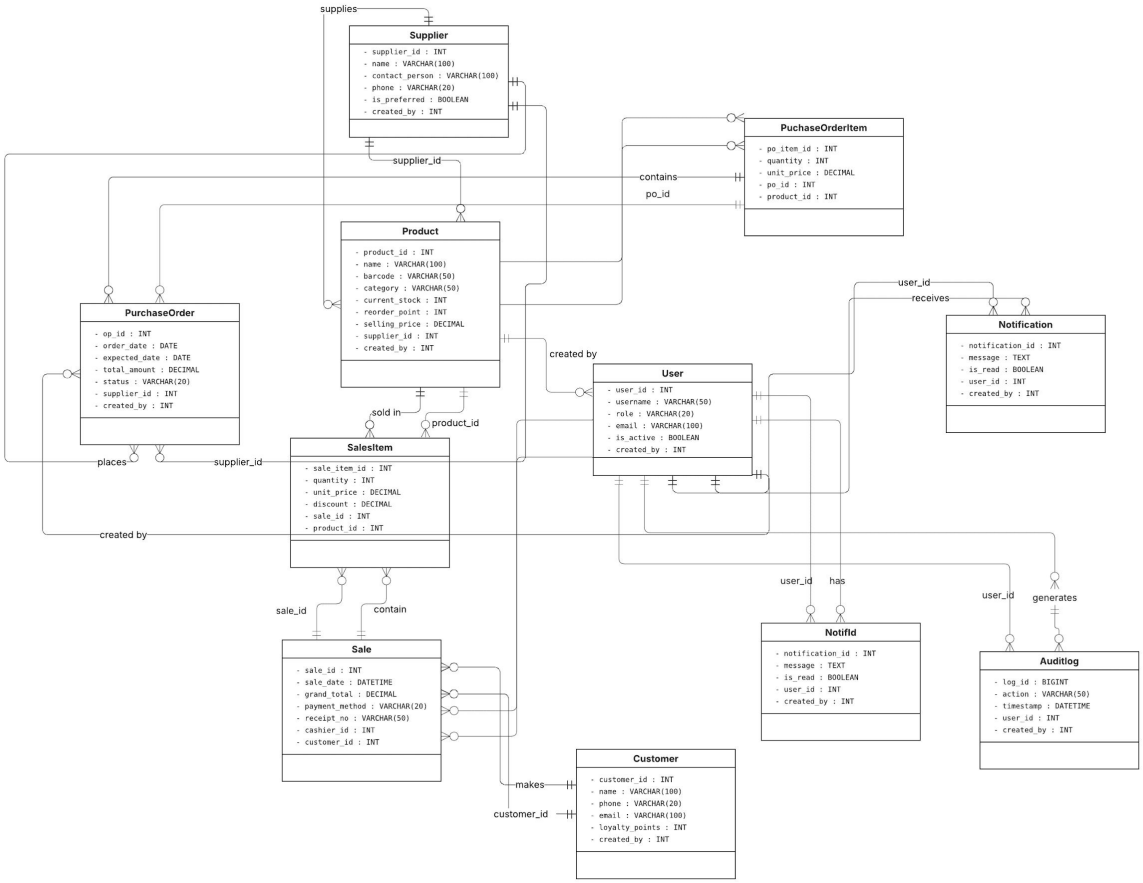


4.7 Class Diagram



4.8 Controller Class Diagram

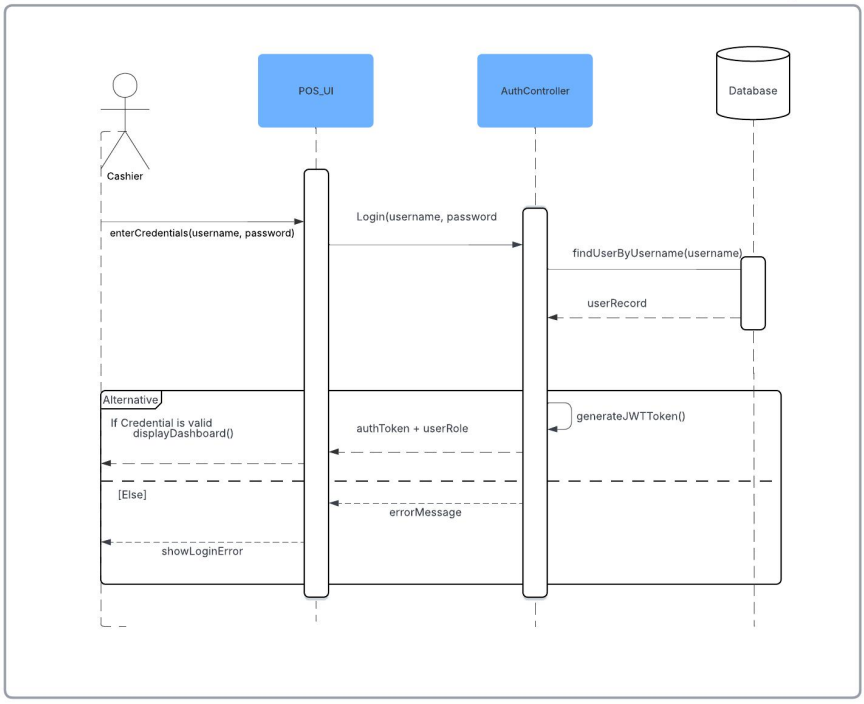
4.8 Entity-relationship (ER) Diagram



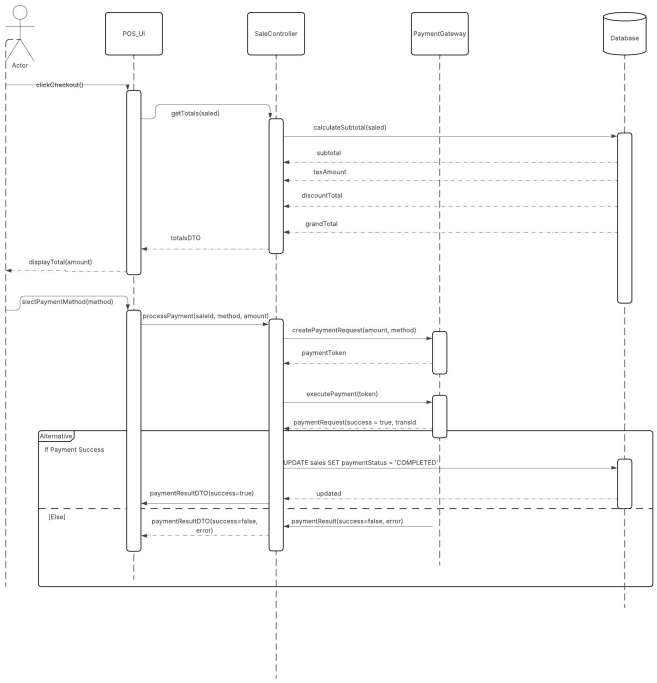
4.9 Sequence Diagram:

Sequence Diagram for Process sale (Divided into 4 parts)

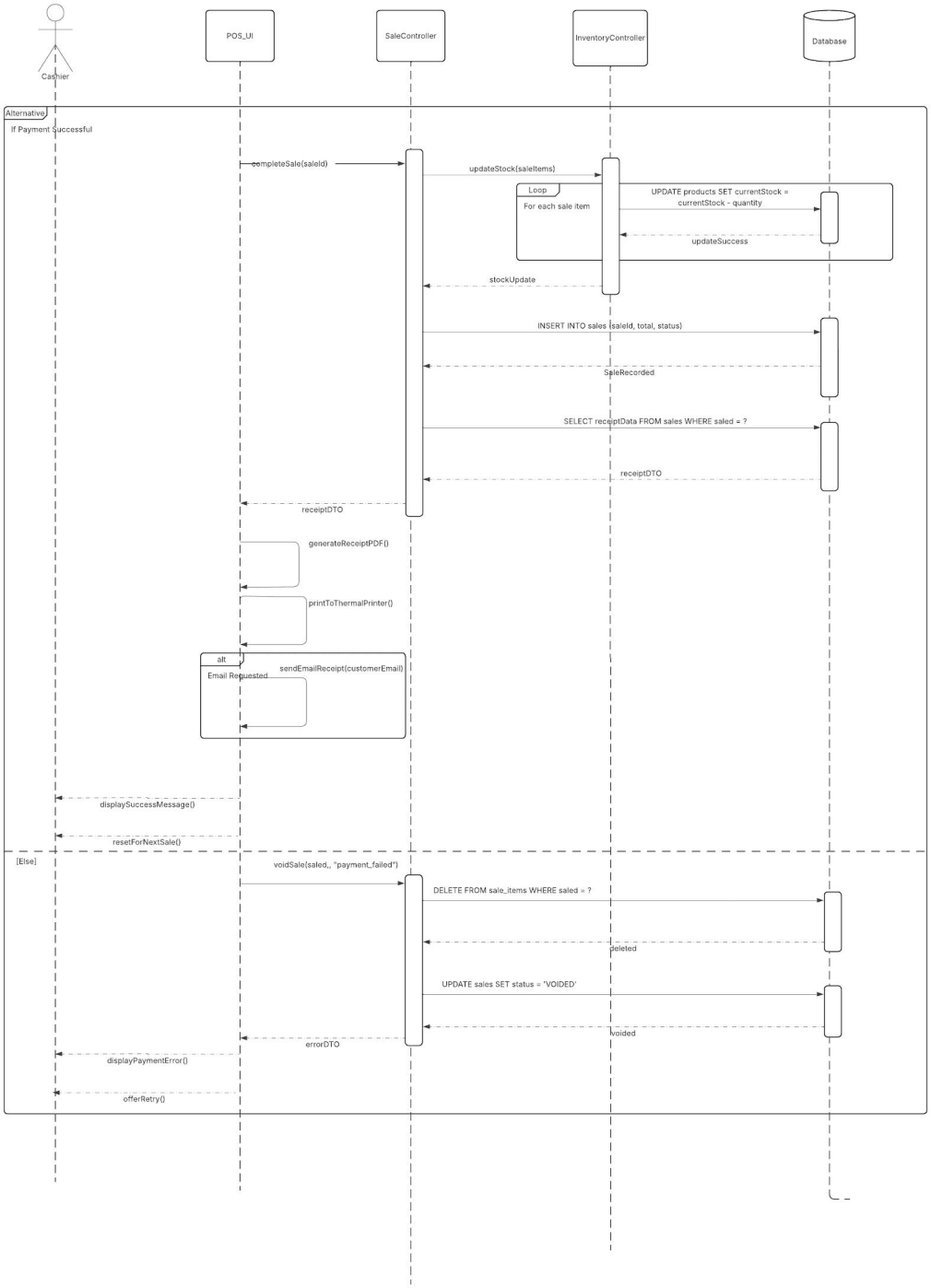
Sequence Diagram Part1/4: Authentication & Login



Sequence Diagram Part3/4: Checkout & Payment Processing



Sequence Diagram Part4/4: Inventory Update & Receipt Generation



5. Requirements Discussion

Functional Requirements:

ID	Requirement
FR1	System shall allow cashier to scan or manually enter product barcode
FR2	System shall calculate total bill including taxes and discounts.
FR3	System shall update inventory in real time after each sale.
FR4	System shall generate lowstock alerts when quantity < reorder point.
FR5	System shall automatically create a purchase order for approved suppliers (optional manual approval). |
FR6	System shall provide daily, weekly, and monthly sales reports.
FR7	System shall support cash, card, and mobile payments via gateway integration.
FR8	System shall maintain customer purchase history .

Non-Functional Requirements:

ID	Requirement	Measurable Criteria
NFR1	Performance	Bill generation < 2 seconds (measured from click "Pay" to receipt ready).
NFR2	Availability	99.5% uptime during business hours (9 AM – 9 PM).

NFR3	Security	Rolebased access (cashier vs owner). All transactions logged. Payment card data tokenized; no PAN stored locally.
NFR4	Usability	Interface requires no more than 3 clicks for common tasks (scan, total, pay). |
NFR5	Scalability	Support up to 5 concurrent POS terminals
NFR6	Offline mode	POS can cache sales locally for up to 2 hours. After 2 hours, system prompts manual sync; cache persists until internet restores – no data loss.

6. Traceability Matrix

Functional Traceability (FR to Use Cases):

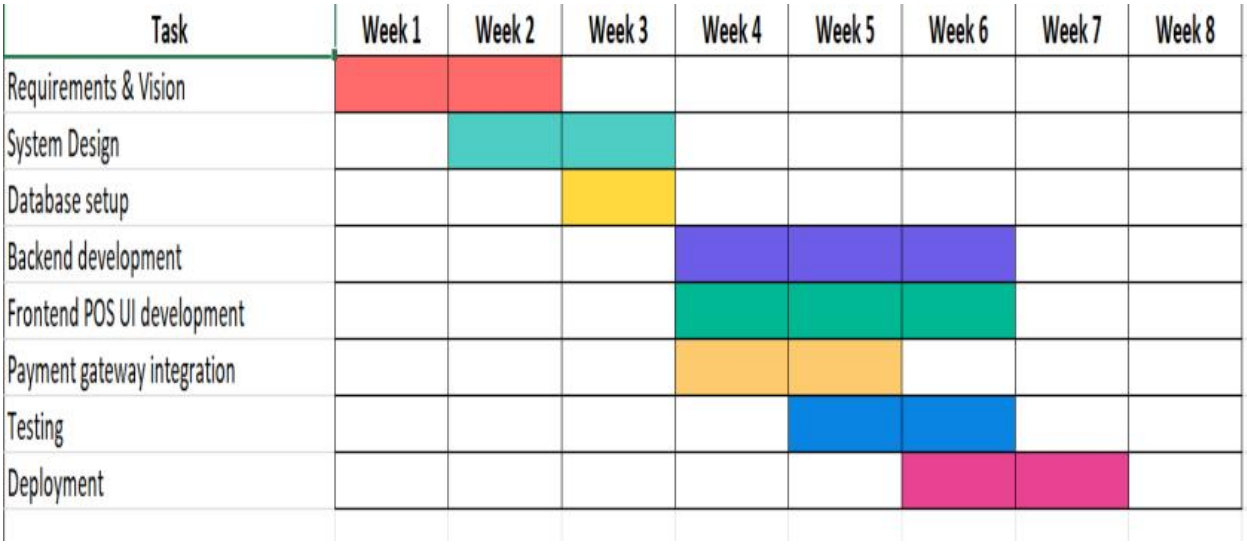
Requirement	UC1: Process Sale	UC2: Manage Inventory	UC3: Auto-Reorder	UC4: Generate Reports
FR1	X			
FR2	X			
FR3	X	X		
FR4		X		
FR5			X	
FR6				X
FR7	X			
FR8	X			

Non-Functional Traceability (NFR to Design Components):

NFR	Design Component / Decision

NFR1 (Performance)	Optimized database indexes on product barcode; inmemory caching for active sale.
NFR2 (Availability)	Cloud hosting with autorestart; database replication.
NFR3 (Security)	JWT tokens for rolebased access; TLS 1.2 for all communications; payment gateway tokenization.
NFR4 (Usability)	Large buttons, barcode shortcut, minimal modal dialogs.
NFR5 (Scalability)	Stateless backend; horizontal scaling possible.
NFR6 (Offline mode)	Local IndexedDB in POS browser; sync queue on reconnection.

7. Project Plan (Gantt Chart)



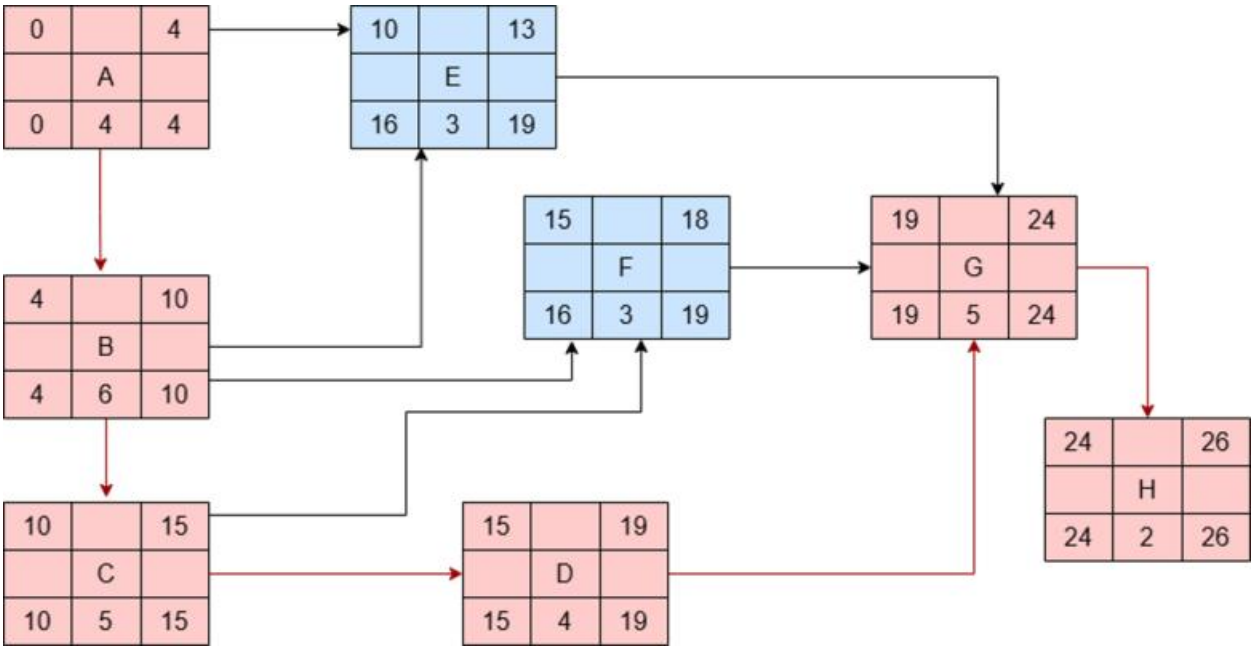
8. Critical Path Analysis

Critical Path:

ID	Description	Predecessor	Duration
A	Requirements	**None**	**4**
B	System Design	**A**	**6**
C	Backend Development	**B**	**5**
D	Frontend Development	**C**	**4**
E	Integration Testing	**A,B**	**3**
F	Deployment	**B,C**	**3**
G	Payment gateway integration	**D,E,F**	**5**
H	Testing and finish	**G**	**2**

Activity	**ES**	**EF**	**LS**	**LF**	**Slack (LS−ES)**	**Critical**
A	0	4	0	4	0	Y
B	4	10	4	10	0	Y
C	10	15	10	15	0	Y
D	15	19	15	19	0	Y
E	10	13	16	19	6	N

F	15	18	16	19	1	N
G	19	24	19	24	0	Y
H	24	26	24	26	0	Y



9. Cost Estimation

Hardware:

Hardware Item	Quantity	Unit Cost (BDT)	Total (BDT)
POS terminal (tablet + stand)	**2**	15,000	30,000
Barcode scanner (USB)	**2**	3,500	7,000
Receipt printer (thermal)	**1**	6,000	6,000
Cash drawer	**1**	4,000	4,000

		Subtotal	**=** 47,000

Software:

Software Item	License / Cost	Total (BDT)
Database (PostgreSQL)	Free	0
Backend (Node.js)	Free	0
Frontend (React)	Free	0
Cloud hosting (AWS Lightsail, 1 year)	1,500/month × 12	18,000
Payment gateway setup fee	Onetime	2,500
Domain & hosting for website (1 year)	2,000	2,000
	****Subtotal****	**=22,500**

Human Resource:

Role	Hours	Rate (BDT/hr)	Total (BDT)
System Analyst	80	500	40,000

Fullstack Developer	200	400	80,000
Tester	60	300	18,000
Trainer (onsite, 1 day)	8	400	3,200
		Subtotal	**= 141,200**

Other (Utility, Rent, etc.):

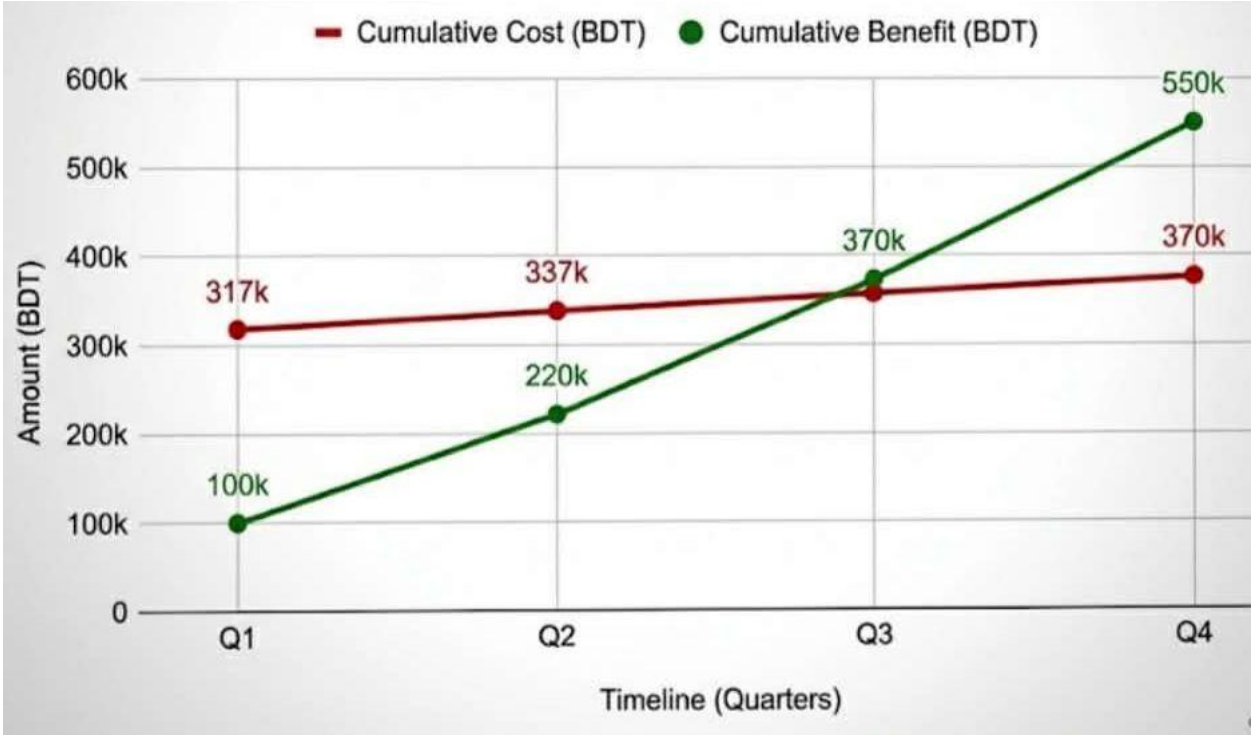
Item	Cost (BDT)
Office rent (2 months for development)	80,000
Utility (electricity, internet for 2 months)	12,000
Licensing (business software permits)	10,000
Miscellaneous (stationery, travel)	5,000
Subtotal	= 107,000

Total Project Cost:

Category	Amount (BDT)
Hardware	47,000
Software	22,500
Human Resource	141,200
Other	107,000
Grand Total	**= 317,700 BDT**

10. Break-Even Point Analysis

Year	Total Cost	Total Benefit	Cumulative Benefit
Quarter 1	317,000	100,000	100,000
Quarter 2	20,000	120,000	220,000
Quarter 3	20,000	150,000	370,000
Quarter 4	20,000	180,000	550,000



11. Risk Register

There are some risks associated with the project that may affect the completion of the work successfully. One of them is data loss when working without internet connection because the system is designed to buffer the sales for two hours and then sync it manually to avoid data loss. The security aspect is also important since the team plans to protect the app using JWT tokens to control role-based access and avoid storing PAN on local devices. In terms of project management, there is a tight timeline, and it is critical path consisting of activities A, B, C, D, G, and H lasts 26 days. Therefore, any delay in the backend and frontend developments (activities C and D) may cause delays in the final submission since there is no float in these activities.

12. Shortcomings and Technical Limitations

Although the system modernizes the processes in the business, it has certain technical constraints. Primarily, the offline feature of the system is limited to just caching in an offline mode and is valid for 2 hours only. Further, scalability is constrained by the number of POS terminals that can be connected to the system, which cannot exceed 5 units. The budget allocated is also limited with regards to the devices provided, as there are just two tablets and one receipt printer. Finally, the fact that the cloud-hosting

platform used in the system is AWS and external payment gateways are used means that the system would face downtime if these are unavailable.

13. Conclusion

The Small Business Technology System offers a complete package that will help small businesses to shift from less efficient methods such as manual systems where record keeping involves papers and order-taking through phone calls. The total amount required for this project to be completed successfully is 317,700 BDT and includes all the necessary components including hardware, software and qualified personnel that will be involved in implementing the system. Using the system in terms of getting inventory reminders and sales reports will make sure that owners make decisions based on accurate information and the customer checkouts will be more convenient.